**[GAME NAME HERE]: An Educational Bubble-shooter game centered around Changes in Matter**

A Capstone Project

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UBAY, Conrad Amadeus Carmelo C.

DIMAUNAHAN, Ryan
Adviser

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## ABSTRACT

Chemistry education contains foundational concepts that are often perceived as difficult for students to understand, particularly in terms of their application to real-world situations. This difficulty may lead to cognitive overload, especially when instruction lacks opportunities for practical engagement. Interactive learning approaches allow students to actively apply and develop their analytical skills while discovering how chemistry concepts relate to everyday experiences.

This study explores the use of Game-Based Learning (GBL) through the development of [GAME NAME HERE], an educational serious game designed as a learning material for Grade 6 students under the MATATAG curriculum in the Philippines. The game focuses on physical and chemical changes of matter, expanding students' understanding beyond the three common states of matter. Through interactive gameplay, students engage with concepts such as phase changes and heat transfer while learning to distinguish physical changes from chemical changes.

The primary goal of this study is to improve students' conceptual understanding, motivation, and emotional engagement by reinforcing hands-on learning and real-world applications of chemistry concepts. This project aims to provide both students and teachers with an accessible and engaging instructional resource that supports visualization, experimentation, and the development of analytical skills in basic chemistry education.

**Keywords:** physical and chemical changes, game-based learning, educational serious game
**Sustainable Development Goals:** 4: Quality Education, 11: Sustainable Cities and Communities, 12: Responsible Consumption and Production

**ABSTRAK**

Ang pag-aaral ng kimika ay naglalaman ng mga pundamental na konsepto na kadalasang itinuturing na mahirap unawain ng mga mag-aaral, lalo na pagdating sa aplikasyon nito sa totoong buhay. Ang ganitong kahirapan ay maaaring magdulot ng cognitive overload, partikular kung kulang ang mga aralin sa praktikal at interaktibong gawain. Sa pamamagitan ng interaktibong pagkatuto, aktibong nahahasa ng mga mag-aaral ang kanilang kasanayan sa pagsusuri at mas nauunawaan nila kung paano nagagamit ang mga konsepto ng kimika sa pang-araw-araw na buhay.

Sinisiyasat ng pag-aaral na ito ang paggamit ng Game-Based Learning (GBL) sa pamamagitan ng pagbuo ng [GAME NAME HERE], isang pang-edukasyong seryosong laro na idinisenyo bilang pantulong na materyal sa pagkatuto para sa mga mag-aaral sa Baitang 6 sa ilalim ng MATATAG kurikulum sa Pilipinas. Nakatuon ang laro sa pisikal at kemikal na pagbabago ng materyal, na nagpapalawak sa kaalaman ng mga mag-aaral lampas sa tatlong karaniwang kaanyuan ng matter. Sa pamamagitan ng interaktibong gameplay, natututuhan ng mga mag-aaral ang mga konsepto tulad ng pagbabago ng estado at paglipat ng init, habang malinaw na natutukoy ang kaibahan ng pisikal at kemikal na pagbabago.

Layunin ng pag-aaral na ito na mapabuti ang pag-unawa, motibasyon, at emosyonal na pakikipag-ugnayan ng mga mag-aaral sa pamamagitan ng hands-on na pagkatuto at aplikasyon ng mga konsepto ng kimika sa mga sitwasyong hango sa totoong buhay. Nilalayon din ng proyektong ito na magsilbing kapaki-pakinabang na materyal para sa mga guro at mag-aaral upang mas mapadali ang pagkatuto ng kimika sa isang biswal, interaktibo, at makabuluhang paraan.

**Mga susing salita:** pisikal at kemikal na pagbabago, game-based learning, pang-edukasyon na laro

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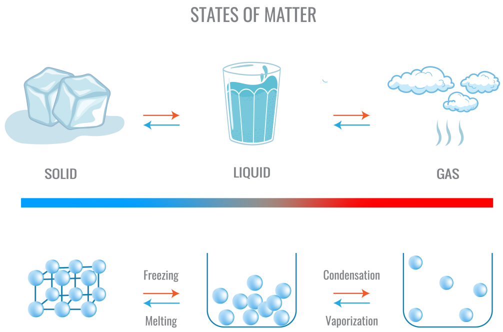
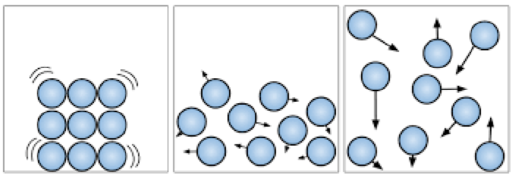
CHAPTER 1
INTRODUCTION

## 1.1 Background of the Study

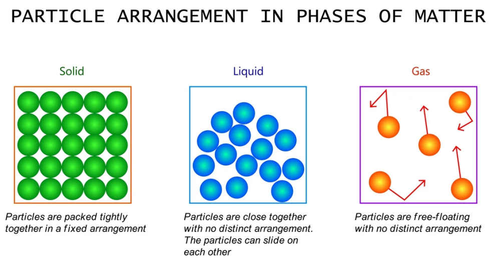
Science education at the elementary level serves as a foundational stage in developing scientific literacy, conceptual reasoning, and problem-solving skills. Within the MATATAG Curriculum of the Department of Education (DepEd), Grade 6 Science introduces learners to fundamental concepts related to materials, including the description of changes of state for solids, liquids, and gases—specifically melting, evaporation, freezing, and condensation—using diagrams and flowcharts. These competencies aim to help learners understand observable phenomena through scientific representations and structured reasoning.




(a) Phase change diagram



(b) Phase changes (Khan Academy)




(c) Phase changes (88Guru)

Figure 1.1: Examples of phase change diagrams used in science education

Despite the importance of these competencies, research has consistently shown that students experience difficulty in understanding the particulate nature of matter and the mechanisms underlying phase changes. Learners often memorize definitions of melting or evaporation without fully comprehending the particle-level interactions that explain these transformations. Misconceptions frequently arise regarding particle spacing, motion, and the role of heat transfer in altering states of matter (Adadan, 2013; Paik, Kim, Cho, & Park, 2004). These challenges are compounded by the abstract nature of microscopic representations, which are not directly observable.

In many classroom settings, demonstrations of phase change are limited to water-based experiments, such as melting ice or boiling water. While these activities provide observable macroscopic changes, they often lack explicit integration with particle-level models, resulting in fragmented conceptual understanding. Furthermore, access to laboratory materials for demonstrating processes such as sublimation or deposition is limited at the elementary level, restricting exposure to diverse examples of state change.

Game-Based Learning (GBL) has emerged as a promising instructional approach that integrates interactive systems with curricular objectives. Educational games can provide dynamic visualizations, immediate feedback, and experiential learning opportunities that reduce cognitive overload and promote conceptual change (Plass, Homer, & Kinzer, 2015). When designed with explicit learning outcomes and aligned assessments, digital games may enhance engagement and deepen understanding, particularly in abstract scientific domains.

This study introduces [GAME NAME HERE], a 2D educational platformer developed as a supplementary and enrichment learning tool aligned with the MATATAG Grade 6 Science competency on changes of state. The game integrates explicit terminology (e.g., "Solid $\rightarrow$ Liquid: Melting") while allowing learners to discover appropriate phase changes through environmental problem-solving. Central to its design is an interactive particle-lattice minigame in which learners manipulate bonds and particle spacing to simulate state transitions. By directly engaging with representations similar to those used in classroom diagrams, learners repeatedly encounter visual models of particle behavior in a contextualized, goal-driven environment.

Given the increasing emphasis on learner-centered instruction under the MATATAG framework, there is a need to explore innovative digital tools that complement formal instruction without replacing it. This study examines whether a platformer-based educational game can support conceptual understanding, engagement, and reinforcement of curriculum-aligned competencies related to phase changes of matter.

## 1.2 Statement of the Problem

This study seeks to investigate the potential of a platformer-based educational game as a supplementary instructional tool for teaching changes of state in Grade 6 Science.

Specifically, the study aims to answer the following questions:

1. To what extent does the use of [GAME NAME HERE] improve learners' conceptual understanding of changes of state (melting, evaporation, freezing, and condensation) at the particle level?

2. How does interaction with the particle-lattice minigame influence students' ability to relate macroscopic phase changes to microscopic particle behavior?

3. What are the perceptions of students and parents regarding engagement, usability, and perceived learning value of the game?

4. How feasible is the implementation of a discovery-based yet explicitly labeled digital game in a supplementary learning context?

## 1.3 Objectives of the Study

The general objective of this study is to design and evaluate a curriculum-aligned educational platformer that reinforces Grade 6 competencies on changes of state.

Specifically, the study aims to:

1. Develop [GAME NAME HERE] as a supplementary digital learning tool aligned with the MATATAG Grade 6 Science competency on describing changes of state using diagrams and flowcharts.
2. Integrate an interactive particle-level simulation that visually represents solids, liquids, and gases and their transitions.
3. Assess learners' conceptual understanding before and after gameplay using structured instruments.
4. Measure student engagement and perceived motivation through survey responses.
5. Collect qualitative insights from students and parents regarding usability and learning reinforcement.

## 1.4 Scope and Delimitations

This study focuses exclusively on the development and evaluation of [GAME NAME HERE], a 2D platformer designed to reinforce the Grade 6 Science competency on changes of state of matter. The game addresses physical changes only and does not include instruction on chemical reactions or chemical change.

The participant pool may include Grade 6 learners as well as students from adjacent grade levels who have previously completed instruction on changes of state. Sampling will follow a convenience-based recruitment model due to logistical constraints. The study adopts a quasi-experimental single-group design incorporating pre- and post-assessments, engagement surveys, and qualitative interviews with students and parents.

The study does not measure long-term retention or academic performance beyond the duration of the intervention. Additionally, the research evaluates the game as a supplementary enrichment tool and does not compare it directly with traditional instruction through a control group.

## 1.5 Significance of the Study

This study contributes to elementary science education by examining the integration of interactive digital modeling within a platformer game environment. For learners, the game provides repeated exposure to particle representations within meaningful problem-solving contexts, potentially strengthening conceptual connections between diagrams and observable phenomena.

For educators, the study offers insights into how educational games may complement MATATAG-aligned instruction without replacing formal teaching. The findings may inform future curriculum integration strategies and digital resource development.

For researchers, the study adds to the growing body of literature on game-based learning in science education, particularly within the Philippine educational context. By examining discovery-based mechanics combined with explicit labeling, the study contributes to ongoing discussions on balancing constructivist learning with guided instruction in digital environments.

## 1.6 Definition of Terms

**Changes of State** – Physical transformations between solid, liquid, and gas phases, including melting, evaporation, freezing, and condensation.

**Particle-Level Representation** – Visual models depicting spacing, motion, and bonding of particles to represent different states of matter.

**Game-Based Learning** – An instructional approach that integrates educational objectives within interactive digital game environments.

**Supplementary Learning Tool** – An instructional resource designed to reinforce and enrich formal classroom instruction without replacing it.

# CHAPTER 2
# REVIEW OF RELATED LITERATURE AND STUDIES

## 2.1 Introduction

This chapter presents the theoretical and empirical foundations relevant to the development of a 2D educational platformer game designed to teach Grade 6 students the concepts of phase changes, heat transfer, and the distinction between physical and chemical changes. The review integrates major learning theories, international research on game-based science education, chemistry-specific digital interventions, and Philippine studies related to educational technology and curriculum alignment.

The chapter is structured into four major sections:

1. Theoretical Foundations
2. International Literature and Studies
3. Local (Philippine) Literature and Studies
4. Synthesis and Research Gap

This review establishes the pedagogical legitimacy and scholarly grounding of the developed educational game.

## 2.2 Theoretical Foundations

### 2.2.1 Conceptual Change Theory

Conceptual Change Theory (Posner et al., 1982) argues that students enter science classrooms with pre-existing mental models that may conflict with scientifically accepted explanations. Learning, therefore, involves restructuring existing conceptions rather than simply adding new information.

For conceptual change to occur, learners must experience:

1. Dissatisfaction with prior conceptions
2. Intelligibility of the new concept
3. Plausibility of the new concept
4. Fruitfulness of the new concept

Research has consistently shown that misconceptions about phase changes are widespread among elementary and middle school students (Driver et al., 1994; Osborne & Cosgrove, 1983). Common misconceptions include:

- Belief that melting alters the substance's identity
- Confusion between evaporation and boiling
- Misunderstanding of energy transfer during state changes
- Belief that physical changes are equivalent to chemical reactions

Interactive visual representations have been found to support conceptual restructuring by explicitly modeling particle behavior during phase transitions (Harrison & Treagust, 2000).

The developed platformer incorporates particle-level manipulation during gameplay. When a player melts an ice block, the lattice structure visually loosens while maintaining particle identity. This directly challenges the misconception that melting changes the substance itself.

Thus, the game design aligns with conceptual change conditions by:

- Creating cognitive conflict (blocked path requiring phase change)
- Providing intelligible visual models (particle rearrangement)
- Demonstrating plausibility (consistent particle identity)
- Reinforcing fruitfulness (applying knowledge across levels)

### 2.2.2 Constructivism

Constructivist theory (Piaget, 1970; Bruner, 1966) posits that knowledge is actively constructed through interaction with the environment. Learning is not transmission but transformation.

In science education, constructivist approaches emphasize:

- Active engagement
- Inquiry-based exploration
- Manipulation of representations
- Iterative hypothesis testing

Digital environments provide rich opportunities for constructivist learning when learners manipulate variables and observe consequences (Jonassen, 1999).

The developed game functions as a constructivist learning space. Players:

- Encounter environmental challenges
- Experiment with phase transitions
- Observe the effects of temperature changes

- Apply learned principles in new contexts

Rather than presenting static diagrams of molecular motion, the game requires learners to actively reorganize particle systems, aligning with constructivist principles of knowledge construction through experience.

### 2.2.3 Multimedia Learning Theory

Mayer's Cognitive Theory of Multimedia Learning (Mayer, 2001; Mayer, 2009) proposes that learning is enhanced when information is presented through coordinated visual and verbal channels.

The theory is based on three assumptions:

1. Dual-channel processing
2. Limited working memory
3. Active processing

Key multimedia principles relevant to the developed game include:

- Spatial contiguity principle
- Temporal contiguity principle
- Coherence principle
- Signaling principle

Research indicates that animation combined with explanatory text improves understanding of dynamic scientific processes compared to text alone (Mayer & Moreno, 2002).

In the developed game:

- Particle animations are synchronized with labels (e.g., "Solid → Liquid: Melting")
- Visual transformations are immediately tied to terminology
- Irrelevant visual elements are minimized during minigames

These design features align with multimedia learning principles and strengthen conceptual integration of scientific language with visual representation.

### 2.2.4 Game-Based Learning Theory

Game-Based Learning (GBL) research emphasizes that games enhance learning when mechanics are directly aligned with instructional objectives (Gee, 2003; Prensky, 2001).

Meta-analyses suggest that digital game-based learning produces moderate positive effects on achievement and motivation in STEM disciplines (Clark et al., 2016; Wouters et al., 2013).

Effective educational games include:

- Clear goals
- Immediate feedback
- Scaffolding
- Incremental difficulty
- Meaningful player agency

The developed platformer integrates:

- Goal-driven progression (reach level endpoint)
- Mechanic-embedded learning (phase change manipulation required to progress)
- Immediate feedback (particle rearrangement animation)
- Increasing complexity across levels

Unlike superficial quiz-based games, the chemical concept is embedded within the core mechanic of progression.

### 2.2.5 Experiential Learning Theory

Kolb's Experiential Learning Theory (Kolb, 1984) conceptualizes learning as a cycle involving:

1. Concrete experience
2. Reflective observation
3. Abstract conceptualization
4. Active experimentation

Game environments are particularly suited to experiential learning because they allow repeated experimentation with immediate consequences (Squire, 2006).

The platformer structure naturally supports the experiential cycle:

- Concrete experience: Encountering a blocked path
- Reflection: Observing failure or transformation
- Conceptualization: Understanding phase change principles
- Experimentation: Applying knowledge in subsequent levels

Through repeated cycles, learners internalize abstract chemical principles.

### 2.2.6 Outcome-Based Teaching and Learning

Outcome-Based Teaching and Learning (OBTL), as articulated by Kalra (2015), emphasizes that educational design should begin with clearly defined learning outcomes, which then determine instructional strategies and assessment methods. Rather than focusing on content delivery alone, OBTL ensures constructive alignment among learning objectives, teaching activities, and assessment practices.

Central principles of OBTL include:

1. Clearly articulated measurable learning outcomes
2. Alignment between outcomes and instructional activities
3. Assessment strategies that directly measure intended competencies

The MATATAG Curriculum of the Department of Education adopts an outcomes-based orientation, emphasizing competency development, conceptual understanding, and application of knowledge.

The developed educational platformer aligns with OBTL principles by:

- Designing gameplay mechanics around specific Grade 6 science competencies
- Embedding phase-change manipulation directly into learning objectives
- Structuring levels to require demonstration of conceptual understanding

Rather than treating the game as supplementary enrichment, the design ensures that progression through levels requires demonstration of intended learning outcomes.

Thus, the game's structure reflects constructive alignment consistent with OBTL principles.

### 2.2.7 Bloom's Taxonomy Framework

Bloom's Taxonomy (Bloom et al., 1956), later revised by Anderson and Krathwohl (2001), categorizes cognitive processes into hierarchical levels:

1. Remember
2. Understand
3. Apply
4. Analyze
5. Evaluate
6. Create

The revised taxonomy emphasizes cognitive processes over static knowledge categories.

In the context of Grade 6 chemistry learning, MATATAG competencies require students not only to recall definitions of phase changes but to understand mechanisms, apply concepts, and distinguish physical from chemical changes.

The developed game supports multiple cognitive levels:

- **Remembering:** Identifying terms such as melting, freezing, evaporation
- **Understanding:** Explaining particle movement during state changes
- **Applying:** Using phase changes to solve in-game obstacles
- **Analyzing:** Distinguishing physical changes from chemical changes

By embedding conceptual problem-solving into gameplay, the platformer moves beyond rote memorization toward higher-order cognitive engagement.

Table 2.4 presents the alignment between Bloom's cognitive levels, MATATAG competencies, and the developed game mechanics.

## 2.3 Theoretical Framework

This study is grounded in established learning theories that explain how learners construct, revise, and apply scientific knowledge. The development of the educational platformer game is primarily anchored in Conceptual Change Theory, Constructivist Learning Theory, and Multimedia Learning Theory, with support from Game-Based Learning Theory, Experiential Learning Theory, and Outcome-Based Teaching and Learning principles aligned with Bloom's Taxonomy. These theoretical foundations collectively inform the instructional design, mechanics, and intended learning outcomes of the intervention.

Table 2.1: Alignment of Bloom's Taxonomy, MATATAG Competencies, and Game Mechanics

| **Bloom's Cognitive Level** | **MATATAG Competency Focus** | **Game Implementation** |
|---|---|---|
| Remember | Identify types of phase changes | UI labels and terminology during minigames |
| Understand | Explain particle behavior during state changes | Animated particle lattice visualization |
| Apply | Use knowledge of heat transfer in context | Manipulating temperature to overcome obstacles |
| Analyze | Distinguish physical from chemical changes | Visual preservation of molecular identity |
| Evaluate (Optional) | Judge appropriate phase change for scenario | Choosing correct state change for progression |

Conceptual Change Theory posits that learners enter the classroom with pre-existing conceptions about scientific phenomena, many of which are misconceptions resistant to change. For conceptual change to occur, learners must experience dissatisfaction with their existing understanding and encounter intelligible, plausible, and fruitful alternative explanations.

In the context of phase changes, learners often hold misconceptions regarding particle motion, energy transfer, and state transitions. The platformer game is designed to confront these misconceptions by allowing learners to manipulate molecular structures directly and observe changes in lattice spacing, particle movement, and energy conditions. By visually and interactively modeling these transitions, the intervention aims to facilitate the restructuring of inaccurate mental models.

Conceptual Change Theory therefore serves as the primary explanatory framework guiding the study's instructional mechanics.

Constructivism asserts that learners actively construct knowledge through interaction with their environment rather than passively receiving information. Learning occurs when individuals engage with tasks, test ideas, and refine understanding through experience.

The game-based intervention reflects constructivist principles by requiring learners to manipulate particle arrangements, experiment with freezing and evaporating mechanics, and solve level-based challenges. Knowledge is not merely presented; instead, learners actively build understanding through interaction and problem-solving.

This theoretical lens justifies the design choice of discovery-based level progression and scaffolded challenges embedded within the platformer environment.

Multimedia Learning Theory suggests that learning improves when information is presented through coordinated visual and verbal channels, provided cognitive processing remains meaningful. Effective multimedia environments integrate text, visuals, and interactive elements to promote deeper understanding.

The platformer incorporates:

- Visual representations of particle movement
- Explicit textual labels (e.g., "Solid → Liquid: Melting")
- Dynamic changes in structure and spacing

These elements are intentionally aligned to reinforce scientific terminology while simultaneously illustrating microscopic processes. The integration of visual simulation and textual labeling supports dual-channel processing and promotes coherent mental model construction.

Game-Based Learning Theory emphasizes motivation, engagement, and structured challenges as mechanisms for promoting learning. The intervention integrates goal-oriented tasks, progressive difficulty, and interactive feedback to sustain learner engagement while targeting specific science competencies.

Experiential Learning Theory proposes that learning occurs through concrete experience, reflective observation, conceptualization, and experimentation. The game enables learners to experience simulated molecular changes, reflect on outcomes, and apply understanding in subsequent levels requiring strategic manipulation of states of matter.

Outcome-Based Teaching and Learning emphasizes alignment between instructional activities and clearly defined learning outcomes. This study aligns game mechanics with competencies specified in the MATATAG Curriculum for Grade 6 Science.

Bloom's Taxonomy further informs the cognitive demands of the game, ensuring that learners move beyond recall toward application and analysis as they manipulate states of matter to solve environmental challenges within the platformer.

Collectively, these theoretical foundations provide the pedagogical justification for the intervention and explain the mechanisms through which conceptual understanding is expected to improve.

### 2.4 Conceptual Framework

This study adopts an Input–Process–Output (IPO) model to illustrate the relationship between theoretical foundations, the game-based intervention, and the intended learning outcomes.

The conceptual framework synthesizes the theories discussed in the previous section and operationalizes them within the context of an educational platformer game designed to teach phase changes of matter.

As illustrated in Figure 2.1, the Input component consists of the theoretical and curricular foundations guiding the study. These include Conceptual Change Theory, Constructivism, Multimedia Learning Theory, Game-Based Learning Theory, Experiential Learning Theory, Outcome-Based Teaching and Learning principles, Bloom's Taxonomy, and the MATATAG Curriculum competencies for Grade 6 Science.


Figure 2.1: Conceptual Framework of the Study

These inputs inform the design of the Process component, which is the educational platformer game. The intervention operationalizes theoretical principles through interactive particle manipulation, phase change simulation mechanics, scaffolded level progression, and explicit labeling of state transitions. Learners actively engage with microscopic representations of matter and apply understanding to overcome in-game challenges.

The Output component represents the intended learning outcomes of the study. These include:

- Improved conceptual understanding of phase changes
- Reduction of misconceptions regarding particle behavior
- Enhanced ability to apply knowledge in problem-solving contexts
- Alignment with targeted MATATAG Grade 6 Science competencies

The arrows connecting the components signify that theoretical and curricular foundations directly shape the instructional design of the intervention, which in turn influences measurable learning outcomes.

This framework guides the development of research instruments and informs the methodology described in Chapter 3.

## 2.5 International Literature and Studies

### 2.5.1 Digital Game-Based Learning in Science Education

International studies have documented positive impacts of digital games on science achievement, engagement, and conceptual understanding (Vogel et al., 2006; Wouters et al., 2013).

A meta-analysis by Clark et al. (2016) found that digital game-based learning had a significant positive effect on cognitive outcomes compared to traditional instruction.

Key findings include:

- Simulation-based games improve understanding of abstract processes
- Interactive environments promote deeper engagement

- Embedded feedback enhances conceptual correction

To synthesize major international findings, Table 2.1 presents selected studies on digital game-based learning in science education.

Table 2.2: Summary of Selected International Studies on Game-Based Science Learning

| **Author(s)** | **Year** | **Subject Area** | **Type of Game/Intervention** | **Key Findings** |
|---|---|---|---|---|
| Vogel et al. | 2006 | Various STEM | Computer games & simulations | Games showed positive effects on cognitive outcomes compared to traditional instruction. |
| Wouters et al. | 2013 | STEM subjects | Serious games | Serious games improved retention and problem-solving skills, especially with feedback. |
| Clark et al. | 2016 | Science & Math | Digital game-based learning | Moderate positive effect size on learning outcomes compared to conventional teaching. |
| Squire | 2006 | Environmental Science | Immersive simulation game | Context-rich gameplay promoted conceptual understanding and systems thinking. |
| Gee | 2003 | General learning | Commercial video games | Effective games align mechanics with learning principles such as feedback and scaffolding. |

### 2.5.2 Educational Games in Chemistry

Research specific to chemistry education emphasizes the importance of connecting macroscopic phenomena with submicroscopic particle models (Johnstone, 1991; Treagust et al., 2003).

Studies show that students struggle to transition between:

- Observable phenomena

- Particle representations
- Symbolic notation

Interactive simulations that allow manipulation of particle systems have been shown to improve conceptual coherence (Harrison & Treagust, 2000).

Games that incorporate submicroscopic modeling are particularly effective in addressing misconceptions about phase transitions and intermolecular forces.

The developed game explicitly integrates particle-level manipulation within a platformer mechanic, thereby linking macro-level environmental change (melting wall) with micro-level representation (lattice loosening).

### 2.5.3 Misconceptions in Phase Changes and Heat Transfer

Extensive research documents misconceptions related to:

- Conservation of matter during state change
- Role of energy in phase transitions
- Distinction between physical and chemical change

Students frequently believe that:

- Boiling destroys matter
- Melting alters chemical composition
- Gas particles disappear

Interactive digital tools have been recommended to address these misconceptions through dynamic visualization (Driver et al., 1994).

The developed platformer directly addresses these documented conceptual difficulties.

## 2.6 Local (Philippine) Literature and Studies

### 2.6.1 Educational Technology in Philippine Science Education

Philippine studies on ICT integration report improved student engagement when interactive tools are incorporated into science instruction (Marcelo, 2012; Caballes, 2016).

However, many implementations involve:

- Multimedia presentations
- Computer-assisted drills

- Video-based instruction

Fewer studies examine immersive digital game environments aligned with curriculum competencies.

### 2.6.2 Game-Based Learning in the Philippine Context

Local research indicates:

- Positive student attitudes toward educational games
- Increased motivation
- Improved short-term retention

Yet gaps remain in:

- Chemistry-specific educational game research
- Particle-level interactive simulations
- Curriculum-aligned digital platformer games

Table 2.2 summarizes selected Philippine studies related to educational technology and game-based learning.

### 2.6.3 Alignment with MATATAG Curriculum

The MATATAG Curriculum emphasizes conceptual understanding and scientific literacy in Grade 6 Science.

Competencies include:

- Explaining changes in state of matter
- Identifying heat transfer in phase transitions
- Distinguishing physical from chemical change

The developed game aligns with these competencies by embedding conceptual application within interactive gameplay.

Table 2.3 presents the alignment between MATATAG Grade 6 Science competencies and the core mechanics of the developed game.

Table 2.3: Summary of Selected Philippine Studies on Educational Technology and Game-Based Learning

| **Author(s)** | **Year** | **Level** | **Subject** | **Key Findings** |
|---|---|---|---|---|
| Marcelo | 2012 | Basic Education | General Science | ICT tools improved engagement but required structured integration for impact. |
| Caballes | 2016 | Secondary | Science | Technology use increased student participation and conceptual retention. |
| Various Local Studies | 2015–2020 | Elementary & Secondary | STEM | Students showed improved motivation; limited chemistry-specific immersive games identified. |

Table 2.4: Alignment of Game Features with MATATAG Grade 6 Science Competencies

| **MATATAG Competency** | **Game Mechanic** | **Intended Learning Outcome** |
|---|---|---|
| Explain changes in state of matter | Particle lattice manipulation minigame | Understand that phase change involves particle rearrangement, not substance change. |
| Describe the role of heat in phase transitions | Temperature adjustment interaction | Recognize heat transfer as the driver of melting, freezing, evaporation. |
| Distinguish physical and chemical changes | Visual preservation of particle identity | Identify that chemical composition remains constant during phase change. |
| Apply understanding of matter in problem-solving | Platform obstacles requiring phase manipulation | Transfer conceptual knowledge to new scenarios. |

### 2.7 Synthesis of Literature

The reviewed literature reveals several consistent findings:

1. Students possess persistent misconceptions regarding phase changes.
2. Particle-level visualization improves conceptual understanding.
3. Game-based learning enhances engagement and cognitive outcomes.
4. Alignment between mechanics and learning objectives is critical.
5. Philippine research lacks immersive chemistry-focused platformer interventions.

Collectively, these findings support the pedagogical rationale for developing an interactive educational platformer grounded in conceptual change and constructivist principles.

### 2.8 Research Gap

Despite strong evidence supporting digital game-based learning in science education, the following gaps remain:

- Limited research integrating submicroscopic particle modeling into platformer mechanics.
- Few curriculum-aligned chemistry educational games designed for Grade 6 learners in the Philippine context.

- Insufficient research examining the impact of interactive lattice manipulation on correcting misconceptions regarding physical versus chemical change.

This study addresses these gaps by developing and evaluating a curriculum-aligned educational platformer embedding conceptual change mechanisms within gameplay.

# CHAPTER 3
# GAME DESIGN

## 3.1 Game Name

The title of the game is [GAME NAME HERE]. The name reflects the game's overarching narrative, which centers on a neighborhood chemist helping community members with everyday challenges. It also incorporates wordplay, as the term "bubbly" refers both to the character's personality and to the game's bubble-shooter mechanics.

## 3.2 Game Overview

[GAME NAME HERE] is a 2D bubble-shooter educational game designed to teach Grade 6 chemistry concepts through interactive gameplay. Players create and transform materials using five action orbs, each representing a specific physical or chemical process. In addition to bubble-shooter levels, players can explore a village environment and assist characters through a sorting minigame focused on identifying salvageable and unsalvageable materials.

## 3.3 Game-Concept

[GAME NAME HERE] is a 2D stylized educational game set in a brightly colored neighborhood and follows a chemist as she assists the community with daily tasks. The challenges focus on core chemistry topics such as states of matter and, more importantly, distinguishing between physical and chemical changes in everyday life. In early levels, particularly those set in a café, players use action orbs on ingredients to observe changes in properties and achieve goals defined by a recipe system.

Progress is communicated through a traffic-light feedback system, where red indicates no progress, yellow indicates partial progress, and green indicates completion. Players must strategically use action orbs such as cooling, heating, mixing, slicing, crushing, and swapping to achieve correct outcomes for each level.

## 3.4 Genre

The game combines educational gaming with puzzle and simulation elements. The bubble-shooter mechanics place it within the puzzle genre, emphasizing logic, accuracy, and planning. The neighborhood exploration component functions as a light simulation, reinforcing learning through contextual storytelling and interaction. This hybrid genre approach supports both active recall and contextual understanding.

### 3.5 Target Audience

The primary target audience of [GAME NAME HERE] is Grade 6 students aligned with the MATATAG curriculum. At this stage, learners are introduced to foundational chemistry concepts such as states of matter, physical and chemical changes, and mixtures. The game's visual style, feedback systems, and mobile-first design make it accessible to a wide range of learners who benefit from interactive and gamified instruction.

### 3.6 Game Flow Summary

The game progresses through increasingly challenging levels, beginning in the café location. As players advance, new ingredients, action orbs, and mechanics are introduced. Bubble-shooter levels demonstrate various physical transformations such as melting, freezing, cutting, and crushing. Sorting minigame levels also increase in complexity by introducing a greater variety of materials.

### 3.7 Look and Feel

The game adopts a cartoon-inspired visual style influenced by titles such as Animal Crossing: New Horizons, Kirby's Dream Buffet, and Hello Kitty Island Adventure. Design inspiration is also drawn from PopMart blind box figures, aligning with the game's whimsical and approachable aesthetic. All assets are 2D, stylized, and rounded to emphasize the game's bubbly theme.

### 3.8 How does the game insert itself in a pedagogical scenario?

[GAME NAME HERE] integrates into pedagogical scenarios by aligning gameplay mechanics with core chemistry learning competencies. It functions as a supplementary instructional tool that reinforces classroom learning through interactive engagement. Teachers may use the game to introduce or review concepts related to physical and chemical changes without the constraints of a physical laboratory. Each level functions as a short lesson paired with a problem-solving activity, providing objectives, challenges, and immediate feedback.

Beyond the bubble-shooter mechanics, the village environment enables contextual application of learned concepts. Players interact with characters, explore locations, and encounter progressively challenging scenarios that connect chemistry concepts to familiar situations. This narrative-driven exploration promotes curiosity and reinforces understanding through storytelling.

The game is designed around two learning objectives aligned with Bloom's Taxonomy.

**LO1 (Lower-Order):** Classify physical changes and chemical changes by identifying states of matter and recognizing whether an action results in a physical or chemical transformation, including examples of uniform and non-uniform mixtures.

**LO2 (Higher-Order):** Explain and justify whether a change is physical or chemical by applying evidence from simulated real-world scenarios, such as predicting outcomes of heating or cooling materials and selecting appropriate actions to achieve desired results.

Through the Village and Bubble Shooter Lab components, [GAME NAME HERE] supports inquiry, experimentation, and critical thinking. Immediate feedback mechanisms, including character reactions and post-level explanations, help correct misconceptions and reinforce accurate understanding.

## 3.9 Gameplay and Mechanics

This section will be talking about the gameplay and mechanics that are going to be shown by the game [GAME NAME HERE]. This section will talk about how the game is played and how the game will supply the necessary learning objectives to the players.

### 3.9.1 Gameplay

Core gameplay takes place in a café setting where players complete chemistry-based challenges focused on changes in states of matter, physical and chemical changes, and the effects of heat on materials. Players launch action orbs such as Heat, Cool, Crush, and Cut toward target materials to trigger appropriate reactions. Incorrect actions result in penalties accompanied by visual and auditory feedback. As players progress, challenges increase in complexity through multi-step reactions and environmental modifiers, with guidance provided to support understanding.

### 3.9.2 Moment to Moment

| **Phase** | **Description** |
|---|---|

Table 3.1: Game Loop Table

### 3.9.3 Mechanics

Set within a narrative-driven environment, players complete tasks inspired by everyday situations, such as food preparation and material sorting. Action orbs simulate scientific processes in an accessible arcade-style format. Progress is visualized through a traffic-light system and a three-star scoring mechanic, enabling players to easily assess their performance.

### 3.9.4 Physics

The game employs basic bubble-shooter physics to simulate predictable trajectories and collisions. These mechanics support intuitive gameplay while reinforcing timing and spatial reasoning skills.

### 3.9.5 Movement

[GAME NAME HERE] uses a top-down perspective for both gameplay and exploration. A virtual joystick enables movement and aiming, allowing seamless control across mobile and desktop platforms.

### 3.9.6 Objects

Interactive objects include action orbs, reaction tiles, collectible tokens, compound-forming elements, and village features such as character kiosks and informational displays. Each object supports learning, progression, or narrative development.

### 3.9.7 Actions

Player actions include aiming and shooting action orbs, switching between available orbs, interacting with characters, unlocking new areas, and viewing educational information within the village.

### 3.9.8 Combat

The game does not include traditional combat. Instead, challenges are presented as puzzles involving resource management, timing, and condition-based problem solving.

### 3.9.9 Economy

Gameplay emphasizes resource management. Players must efficiently use limited material orbs to complete level objectives, encouraging critical thinking and strategic planning.

### 3.9.10 Game Options

[GAME NAME HERE] includes adjustable sound and control sensitivity settings, a contextual help menu, and a pause menu. These options enhance usability and player comfort without interfering with educational objectives.

#### 3.9.11 Constraints due to the pedagogical objectives

To maintain pedagogical integrity, gameplay mechanics are intentionally simplified and scaffolded. All reactions align with Grade 6 learning outcomes, and pacing is designed to minimize stress while reinforcing understanding.

### 3.10 Story, Setting and Character

#### 3.10.1 Story and Narrative

The narrative follows a chemist assisting in the opening of a neighborhood café. Through completing recipes and tasks, the player observes how ingredients undergo physical and chemical changes. Later challenges introduce irreversible chemical changes, reinforcing key distinctions between transformation types and encouraging reflection on real-world applications.

#### 3.10.2 Game progression

Progression is structured to introduce new concepts gradually, allowing players time to practice and internalize information before advancing. Narrative and gameplay progression are designed to support continuous learning without cognitive overload.

#### 3.10.3 License considerations

All third-party and open-source assets will be properly licensed and credited. In-house code will follow MIT licensing, while creative assets will use Creative Commons Attribution-NonCommercial 4.0 (CC BY-NC 4.0).

#### 3.10.4 Game World

The game world centers on a chemistry-themed café presented in a stylized 2D environment. The setting blends cozy design with experimental elements, providing a playful space for discovery, experimentation, and problem solving.

#### 3.10.5 Characters

Players assume the role of a research assistant guided by the Local Bubble Chemist. Through completing tasks and experiments, players unlock new mechanics and increasingly complex challenges.

#### 3.10.6 Levels

Levels follow a classic bubble-shooter structure inspired by games such as Zuma, combined with exploration and interaction reminiscent of life-simulation games. Friendly guidance and narrative context support learning progression.

### 3.10.7 Training levels

The training level introduces the game's mechanics and educational goals in a low-pressure environment. Visual cues, simple prompts, and immediate feedback build player confidence and prepare learners for more complex challenges.

### 3.10.8 Assessment

Player analytics will be integrated to track engagement and learning-related metrics such as playtime and level completion. These data will support analysis of learning progress and engagement.

## 3.11 Interface

### 3.11.1 Visual System

A top-down view ensures clear visibility of the play area. Interface elements are minimal and designed to present essential information without obstructing gameplay.

### 3.11.2 Control System

The game uses screen-based controls optimized for both mouse and touchscreen input, ensuring accessibility across platforms.

### 3.11.3 Audio, music, sound effects

Audio assets will be sourced from in-house production, open-source libraries, and licensed third-party resources, with proper attribution provided.

### 3.11.4 Help System

The in-game help system provides tutorials, definitions, and visual examples accessible from the main menu and pause screen, allowing learners to review concepts at their own pace.

# CHAPTER 4
# DESIGN AND IMPLEMENTATION

## 4.1 Design Changes

## 4.2 Implementation Challenges

## 4.3 Formative User Testing Observations

Formative user testing was conducted with Grade 6 students from Inchican Elementary School using a playable prototype that included the Tutorial Level, Bubble Shooter Level 1, and the Sorting Mini Game. Testing was conducted face-to-face, allowing direct observation of player behavior, engagement, and comprehension. Students demonstrated an overall understanding of gameplay mechanics and responded positively to visual cues and immediate feedback, despite their current science curriculum focusing on topics outside states of matter.

Feedback indicated that some levels were perceived as too easy and quickly memorized, suggesting a need for increased challenge variety. Students also commented on improving visual consistency in the sorting minigame and adding more objects to extend engagement. Language accessibility emerged as a key consideration, with requests for Filipino or Tagalog text. Overall, students expressed enjoyment of the game experience and interest in playing a completed version. These observations will inform future refinements to difficulty balancing, visual design, and language support.

# CHAPTER 5
# METHODOLOGY

## 5.1 Research Design

This study employs a mixed-methods research approach. Qualitative data will be collected through one-on-one interviews conducted during and after game development to identify student preferences and evaluate enjoyment. Quantitative data will be gathered through pretests and posttests administered over one month to assess learning gains related to changes in the state of matter.

## 5.2 Data Collection

Primary data will be collected from interviews and pretest and posttest results involving 20 Grade 6 students in a face-to-face setting. Each researcher will interview five students, with interviews audio-recorded for transcription and analysis. Pretests will assess baseline knowledge through a 25-item quiz administered prior to gameplay. Posttests of equal length will be administered following gameplay sessions monitored weekly by the research team.

Secondary data will include in-game analytics designed to track engagement and progression. These data will be collected ethically, stored securely, and used to supplement findings from tests and interviews.

## 5.3 Data Analysis

Repeated measures analysis of variance (ANOVA) will be used to compare pretest and posttest scores (DataTab Team, 2025). Qualitative interview data will be analyzed using thematic analysis, involving transcription, coding, theme identification, and interpretation to synthesize findings (Hecker & Kalpokas, 2025).

## 5.4 Ethical Considerations

Informed consent will be obtained from all participants and their parents or guardians. Participant anonymity will be maintained, and students may withdraw at any time without penalty. Potential risks, such as content misalignment with classroom instruction, will be mitigated through collaboration with teachers and iterative playtesting. All participant data will be securely disposed of upon completion of the study.

# REFERENCES

DataTab Team. (2025). Repeated measures anova. https://datatab.net/tutorial/anova-with-repeated-measures

Hecker, J., & Kalpokas, N. (2025, February). Thematic analysis in mixed methods approach. https://atlasti.com/guides/thematic-analysis/thematic-analysis-mixed-methods